Example 5.1

$$X_t = \frac{1}{2} X_{t-1} + Z_t - \frac{3}{10} Z_{t-1}$$

(a) we have:

$$\left(1 - \frac{B}{2}\right) X_t = \left(1 - \frac{3}{10}B\right) Z_t$$

$$\phi(B) X_t = \theta(B) Z_t$$

where $\phi(B) = 1 - \frac{B}{2}$ and $\theta(B) = 1 - \frac{3}{10}B$

The process is (weakly) stationary if the roots of $\phi(z) = 0$ lie outside the unit circle

$$1 - \frac{z}{2} = 0 \Rightarrow z = 2 > 1 \quad \therefore \text{(weakly) stationary.}$$

The process is invertible if the roots of $\theta(z) = 0$ lie outside the unit circle.

$$1 - \frac{3}{10}z = 0 \Rightarrow z = \frac{10}{3} > 1 \quad \therefore \text{invertible}$$

(b)
$$X_t = \frac{\theta(B)}{\phi(B)} Z_t$$

$$= \frac{\left(1 - \frac{3}{10}B\right)}{\left(1 - \frac{B}{2}\right)} Z_t$$

$$= \left(1 - \frac{3}{10}B\right)\left(1 - \frac{B}{2}\right)^{-1} Z_t$$

Using a Taylor series expansion

$$\left(1 - \frac{B}{2}\right)^{-1} = \sum_{j=0}^{\infty} \left(\frac{1}{2}\right)^j B^j$$

$$\therefore X_t = \left(1 - \frac{3B}{10}\right) \sum_{j=0}^{\infty} \left(\frac{1}{2}\right)^j B^j Z_t$$

$$= \sum_{j=0}^{\infty} \left(\frac{1}{2}\right)^j B^j Z_t - \frac{3}{10} \sum_{j=0}^{\infty} \left(\frac{1}{2}\right) B^{j+1} Z_t$$

$$= Z_t + \sum_{j=1}^{\infty} \left(\frac{1}{2}\right)^j B^j Z_t$$

$$- \frac{3}{10} \sum_{j=0}^{\infty} \left(\frac{1}{2}\right)^j B^{j+1} Z_t$$

$$= Z_t + \left(\frac{1}{2} - \frac{3}{10}\right) \sum_{j=1}^{\infty} \left(\frac{1}{2}\right)^{j-1} Z_{t-j}$$

$$= Z_t + \frac{1}{5} \sum_{j=1}^{\infty} \left(\frac{1}{2}\right)^{j-1} Z_{t-j}$$

(c) $$Z_t = \frac{\phi(B) X_t}{\theta(B)}$$

$$= \frac{\left(1 - \frac{B}{2}\right) X_t}{\left(1 - \frac{3B}{10}\right)}$$

$$= \left(1 - \frac{B}{2}\right)\left(1 - \frac{3}{10}B\right)^{-1} X_t$$

$$\left(1 - \frac{3}{10}B\right)^{-1} = \sum_{j=0}^{\infty} \left(\frac{3}{10}\right)^j B^j$$

$$\therefore \quad Z_t = \left(1 - \frac{B}{2}\right) \sum_{j=0}^{\infty} \left(\frac{3}{10}\right)^j B^j X_t$$

$$Z_t = X_t + \left(\frac{3}{10} - \frac{1}{2}\right) \sum_{j=1}^{\infty} \left(\frac{3}{10}\right)^{j-1} X_{t-j}$$

$$= X_t - \frac{1}{5} \sum_{j=1}^{\infty} \left(\frac{3}{10}\right)^{j-1} X_{t-j}$$

$$\therefore \quad X_t = \frac{1}{5} \sum_{j=1}^{\infty} \left(\frac{3}{10}\right)^{j-1} X_{t-j} + Z_t \, .$$